

Project Description

Title :

Resume Screening & Candidate Ranking System using NLP

Objective :

Build an AI-powered system that can analyze resumes and rank candidates based on a job description using Natural Language Processing (NLP).

Project Overview :

The interns will create a mini AI recruitment assistant where:

- User uploads multi-formate resumes (PDF/DOCX/TXT)
- User enters a job description
- AI extracts text from resumes
- NLP model compares resumes with the job description
- System ranks candidates based on relevance score
- Dashboard displays top candidates

Expected Features :

Core Features (Mandatory)

1. Resume Parsing

Extract text from:

- PDF resumes
- DOCX resumes

Libraries:

- PyPDF2
- pdfplumber
- python-docx

2. Text Preprocessing

Perform:

- Lowercasing
- Stopword removal
- Tokenization
- Lemmatization

Libraries:

- NLTK
- spaCy

3. Similarity Matching

Compare:

- Resume text
- Job description text

Recommended:

- TF-IDF + Cosine Similarity

Advanced (Bonus):

- Sentence Transformers / BERT embeddings

4. Candidate Ranking

Generate:

- Match percentage
- Top skills matched
- Ranked candidate list

Example:

Rank	Candidate	Match Score
1	Rahul Sharma	89%
2	Priya Patel	82%

5. Simple Frontend/UI

Can use:

- Streamlit (recommended)
or
- Flask + HTML
- or any UI according to you..

UI should allow:

- Upload resumes
- Paste job description
- Show rankings

Deliverables

At the end of 1 week they should submit:

Mandatory

- Working application
- GitHub repository
- README documentation
- Requirements.txt

Suggested Tech Stack (used according to you skills)

Area	Technology
Language	Python
NLP	NLTK / spaCy
ML	Scikit-learn
UI	Streamlit + React + other libraries
File Parsing	PyPDF2, pdfplumber
Deployment (Optional)	Hugging Face / Render

Dataset & Resources (reference only used any):

Resume Dataset :

Kaggle Resume Dataset - [Resume Dataset \(Kaggle\)](#)

Updated Resume Dataset - [Resume Screening Dataset](#)

Learning References :

NLP Basics -

[NLTK Documentation](#)

[spaCy Documentation](#)

TF-IDF + Cosine Similarity -

[Scikit-learn TF-IDF Tutorial](#)

[Cosine Similarity Documentation](#)

Streamlit -

[Streamlit Official Docs](#)

Note :

- The references and steps in this document are provided only to help you understand the requirements. You can use any resources, tools, or approach that works best for you.
- These references are shared to save your research time and give you a starting point. Our main expectation is a fully working application that meets all requirements using your own development process and logic.